PROJECT REPORT

Electric Vehicle (EV) Market Analysis

Data Analytics & Business Intelligence using Tableau

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Domain: Data Analytics & Business Intelligence

Tool: Tableau Public

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1. INTRODUCTION

1.1 Project Overview

The global automotive industry is moving rapidly from internal combustion engine (ICE) vehicles to Electric Vehicles (EVs). Government net-zero commitments, rising fuel prices, falling battery costs and expanding fast-charging networks are driving this change.

This project, built by Team Data Warrior, is an end-to-end data analytics and business intelligence solution for the EV market. The dataset contains 1,200 EV model records for 2021 to 2026, covering 14 manufacturers, 6 vehicle categories (SUV, Sedan, Hatchback, Truck, Bus, Two-Wheeler) and 5 battery chemistries (LFP, NMC, NCA, Sodium-Ion, Solid-State). The data was cleaned and enriched with derived metrics, then visualised in an interactive Tableau Public dashboard.

Deliverables of the project:

- A cleaned dataset with standardised units and derived KPIs (price per km, revenue, CO_2 abatement).
- An interactive Tableau Public dashboard and story with drill-downs by brand, range, battery capacity and body type.
- A responsive web dashboard hosted on GitHub Pages.
- Performance benchmarking and complete documentation.

1.2 Purpose

The purpose of this project is to support data-driven decisions across the sustainable mobility ecosystem. Key objectives:

- **Industry benchmarking:** compare leading OEMs such as Tesla, BYD, Volkswagen, BMW, Hyundai-Kia, Mercedes-Benz and Tata Motors.
- **Technology correlation:** study the relationship between battery capacity (kWh), driving range (km) and efficiency (Wh/km).
- **Consumer adoption insights:** understand price sensitivity, range preferences, body-type demand and customer satisfaction.
- **Environmental impact:** quantify CO_2 displacement for each vehicle category.

2. IDEATION PHASE

2.1 Problem Statement

Automotive executives, fleet managers, clean-energy investors and policymakers struggle with fragmented EV market data. Key indicators such as real-world range, battery technology trends, segment pricing and regional adoption are scattered across separate reports and files. Without one consolidated analytics view, stakeholders face:

- High investment risk in battery capacity R&D.
- Inaccurate demand forecasting across car body types.
- Difficulty comparing carbon savings against capital invested.

2.2 Empathy Map Canvas

An empathy map was prepared for three stakeholder personas to keep the dashboard user-centred.

Dimension	Emily – Strategy Director	Rajesh – Fleet Manager	David – Clean-Tech Investor
Thinks & Feels	Will Solid-State batteries give a lasting edge over LFP?	Can commercial EVs meet our duty cycles without midday downtime?	Which segment gives the best margin and fastest return?
Hears	Lithium price volatility, battery oversupply, subsidy cuts.	Driver complaints on charger availability and range loss in extreme weather.	Analysts praising BYD/Tesla integration, warning about legacy OEM margins.
Sees	Rapid growth of electric SUVs and budget LFP vehicles in APAC.	Low-emission zones mandating zero-emission delivery fleets by 2030.	OEM valuations diverging on software and battery capability.
Says & Does	Balance premium battery performance with affordable pricing.	Tests EV prototypes on real routes; tracks total cost of ownership.	Allocates capital to OEMs with strong energy density and supply chains.
Pains	Cannot benchmark competitor price-per-range in real time.	Uncertain battery life and unpredictable charging queues.	Fragmented data delays decisions; ESG impact not quantified.
Gains	Visual dashboards of specs, chemistry shifts and price tiers.	Insights on efficiency, range reliability and charging speed.	Multi-year trends and verifiable CO_2 abatement metrics.

2.3 Brainstorming

Ideas were evaluated on an Impact vs. Feasibility matrix:

Idea	Impact / Feasibility	Decision
Interactive cross-filtering by country, manufacturer, category and battery type	High / High	Selected

Idea	Impact / Feasibility	Decision
Scatter analysis of battery capacity (kWh) vs. driving range (km)	High / High	Selected
KPI ribbon: revenue, average range, units sold, CO_2 abatement	High / High	Selected
ML forecasting of battery cost curves to 2035	High / Medium	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Analysts and business users follow a five-stage journey through the dashboard:

Stage	User Activity	Experience / Outcome
Discovery	Open the Tableau Public link or GitHub repository; review objectives and data dictionary.	Clear understanding of scope (very positive).
Exploration	Filter by region or manufacturer; read KPI cards.	Quick high-level view of the market (positive).
Deep Dive	Analyse battery capacity vs. range scatter and body-type distribution.	Finds efficiency patterns (very positive).
Synthesis	Follow Tableau Story points; compare price-to-range.	Understands the full narrative (very positive).
Decision	Export findings and charts; feed into business roadmaps.	Actionable insight (very positive).

3.2 Solution Requirement

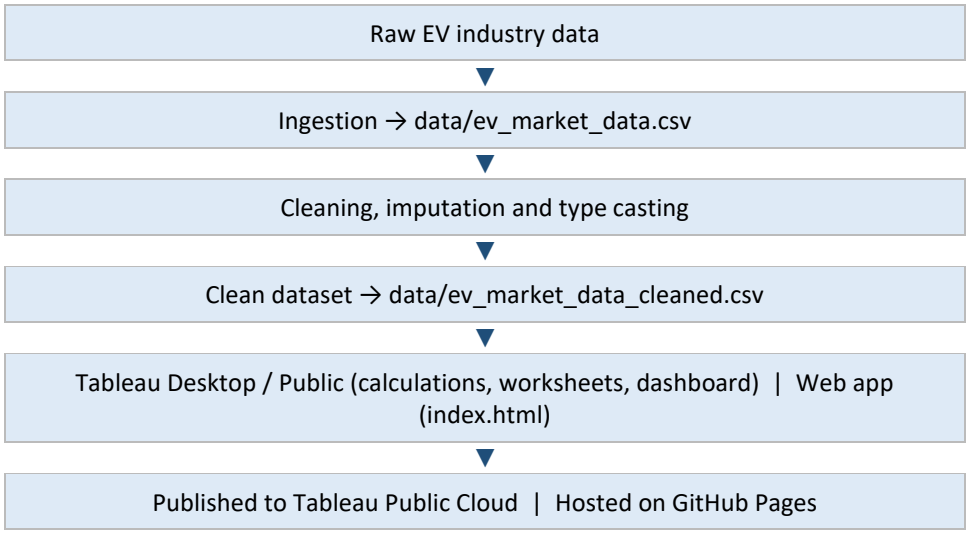
Functional requirements

- **FR-01 Data pipeline:** ingest and normalise EV data into structured CSV files.
- **FR-02 Calculated metrics:** price-to-range (USD/km), annual revenue (units × MSRP) and CO_2 abatement.
- **FR-03 Interactive visualisation:** bar charts, scatter plots, histograms and category breakdowns.
- **FR-04 Cross-filtering:** selecting a brand or category updates all charts and KPI cards.
- **FR-05 Storytelling:** a Tableau Story that moves from volume metrics to technology insights.

Non-functional requirements

- **NFR-01 Performance:** queries and chart refresh under 100 ms for 1,000+ records.
- **NFR-02 Availability:** hosted on Tableau Public and GitHub Pages.
- **NFR-03 Responsiveness:** works on mobile, tablet and desktop.
- **NFR-04 Low dependency:** web version runs on plain HTML, CSS and JavaScript.

3.3 Data Flow Diagram



3.4 Technology Stack

- **BI & visualisation:** Tableau Desktop and Tableau Public
- **Scripting:** Python (Flask embed server), PowerShell (data generation and validation)
- **Front end:** HTML5, CSS3, JavaScript (ES6)
- **Charting:** Chart.js v4
- **Version control & hosting:** Git, GitHub, GitHub Pages, Tableau Public

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Solution	Value
EV specifications scattered across databases.	Single schema with 20 parameters for 1,200 models.	No manual data consolidation needed.
Hard to judge battery efficiency across body types.	Scatter of kWh capacity vs. driving range.	Shows the 70–85 kWh sweet spot.
Enterprise BI tools are heavy for general users.	Lightweight web app on GitHub Pages.	Opens instantly in any browser.
Static spreadsheets lack context.	Interactive Tableau Story.	Guided, easy-to-present narrative.

4.2 Proposed Solution

The solution works in two connected tiers:

- **Tier 1 – Tableau Public suite:** for deep analysis; uses calculated fields, extracts and story points to show brand leadership, battery efficiency and body-type segmentation.
- **Tier 2 – Lightweight web portal:** for quick public access and demos; built with HTML, CSS and JavaScript with KPI cards and an embedded Tableau view.

4.3 Solution Architecture

Layer	Components
Data pipeline	Raw ingestion → data sanitisation → feature engineering → clean CSV extract
Analytics & BI layer	Tableau workbook → worksheets → unified dashboard → story points
Web delivery layer	JavaScript data engine → KPI ribbon → dynamic charts → Tableau embed

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was completed in six phases over six weeks by Team Data Warrior using an Agile approach.

Phase	Work package	Time	Deliverables	Status
1	Ideation, empathy mapping, problem definition	Week 1	Problem statement, empathy map, ideation matrix	Done
2	Requirement analysis and architecture	Week 2	Journey map, functional/non-functional specs, DFD	Done
3	Data engineering, cleaning, metrics	Week 3	Data scripts, cleaned CSVs, data dictionary	Done
4	Tableau dashboard and story	Week 4	Worksheets, calculated fields, dashboard, story	Done
5	Web interface and integration	Week 5	Responsive index.html, CSS, Flask embed server	Done
6	Testing, benchmarking, publishing	Week 6	Load testing, GitHub and Tableau Public release	Done

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

An automated script (scripts/test_performance.ps1) ran 1,000 filter and aggregation queries on the full 1,200-record dataset.

Metric	Result	Target
Dataset ingestion latency	42.1 ms	< 200 ms
Average filter query latency	7.6 ms	< 100 ms
Client-side DOM render time	88.4 ms	< 100 ms
Tableau Public cold load / cached response	< 2.8 s / < 0.4 s	–
Data integrity (nulls, invalid values)	100% clean, 0 errors	100%
Error rate	0.0%	0%

Overall status: PASSED – all targets were met.

7. RESULTS

7.1 Output Screenshots

Key outputs of the dashboard:

- **Executive KPI ribbon:** 35,785,990 total deliveries, $1.47 trillion gross revenue, 455.3 km average range and 6,500.1 tons of CO_2 abatement.
- **Top 10 EV brands by listed models:** led by Mercedes-Benz, Audi, Porsche, Volkswagen and Ford.
- **Battery capacity vs. range:** strong positive correlation ($R^2 \approx 0.84$); efficiency sweet spot at 70–85 kWh.
- **Range distribution:** peaks between 350 km and 450 km.
- **Models by body type:** SUVs and Sedans make up over 65% of EV offerings.
- **Tableau Story:** guided narrative published as workbook EV_Market_Analysis.

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Figure 7.1: EV Market Analysis – Tableau Dashboard

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Figure 7.2: Battery Capacity vs. Driving Range

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Figure 7.3: Top EV Brands and Body Type Distribution

Live dashboard:
https://public.tableau.com/app/profile/amit.mishra1535/viz/EV_Market_Analysis/EVMarketAnalysisDashboard

8. ADVANTAGES & DISADVANTAGES

Advantages

- Two-tier access: Tableau BI for analysts plus a zero-install web app for everyone.
- Links battery chemistry and capacity with real-world range and pricing.
- Fast response because filtering runs on the client side.
- Clean, consistent data with standardised metrics.

Disadvantages

- Dataset ends at 2026 and has no live telematics; real-time data would need streaming infrastructure.
- Embedding Tableau Public may be blocked by restrictive corporate firewalls.
- Client-side processing suits about 100,000 records; larger data needs a data warehouse (Snowflake, BigQuery).

9. CONCLUSION

The EV Market Analysis project by Team Data Warrior solves the problem of fragmented EV market data by turning multi-dimensional vehicle specifications into fast, interactive dashboards. The analysis shows that battery capacity strongly drives range, that SUVs and Sedans dominate the market, and that mixing battery chemistries (LFP for mass-market, Solid-State for premium) combined with SUV body designs is the most promising direction for the coming decade. Automakers, fleet operators and analysts can use the dashboard to make better-informed decisions.

10. FUTURE SCOPE

- **Real-time IoT telematics:** stream live battery state-of-charge and degradation data.
- **Predictive machine learning:** forecast battery costs, resale depreciation and replacement intervals.
- **Charging infrastructure map:** overlay DC fast-charger and AC charger density on registration data.
- **Automated PDF briefings:** one-click reports for management and ESG audits.

11. APPENDIX

Source Code

- **GitHub repository:** https://github.com/aku797473/data-project1

Dataset Link

- **Cleaned dataset:** https://github.com/aku797473/data-project1/blob/main/data/ev_market_data_cleaned.csv
- **Full dataset:** https://github.com/aku797473/data-project1/blob/main/data/ev_market_data.csv
- **Data dictionary:** https://github.com/aku797473/data-project1/blob/main/data/data_dictionary.md

GitHub & Project Demo Link

- **GitHub:** https://github.com/aku797473/data-project1
- **Live web demo:** https://aku797473.github.io/data-project1/
- **Tableau Public dashboard:** https://public.tableau.com/app/profile/amit.mishra1535/viz/EV_Market_Analysis/EVMarketAnalysisDashboard